Intergenerational Income Mobility and Private Education in South Korea between 1998–2024

Step 1 : Descriptive Analysis of Long-Run Trends (Birth Cohorts 1960–2000)

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Abstract

This study systematically analyzes intergenerational income mobility across birth cohorts from 1960 to 2000, using waves 1 through 27 (1998–2024) of the Korean Labor and Income Panel Study (KLIPS). Using a rigorously deduplicated sample of 5,388 unique children (one observation per child), the baseline intergenerational income elasticity (IGE) is estimated at **0.131** (individual-level clustered SE = 0.013), rising to 0.178 when controlling for parental age and birth cohort. The rank–rank slope is 0.163. Cohort-level analysis reveals an apparent decline in IGE from approximately 0.25 (1960s cohort) to approximately 0.14 (1990s cohort), though both ends of this trend are affected by lifecycle bias — upward bias for early cohorts (parental income observed at ages 55–65) and downward bias for late cohorts (children observed at ages 25–30) — and the pattern is interpreted as suggestive rather than structural. For the lifecycle-adjusted core sample (cohorts 1970–1994), the IGE is 0.154. Daughters exhibit approximately twice the IGE of sons, a finding overlooked by prior Korean literature focused on sons. Quantile regression indicates that intergenerational persistence is strongest at the bottom of the distribution (10th percentile IGE 0.161 vs. 90th percentile 0.057). Regional analysis confirms a Capital Region > Metropolitan > Province hierarchy in intergenerational persistence. Across income definitions and sample restrictions, the IGE ranges from 0.056 to 0.229; cross-national comparisons are presented as a range rather than a single figure. These results substantially extend Kwon and Jeon (2020)'s analysis of the 1982–1990 cohort to the full 1960–2000 range and provide the descriptive foundation for the multilevel analysis (HLM) and movers design in Step 2.

1 Introduction

1.1 Background

Intergenerational income mobility is a core indicator of equality of opportunity in a society. As the Great Gatsby Curve (GGC) demonstrates, there is a positive relationship between cross-sectional income inequality and the intergenerational persistence of economic status (Corak 2013; Krueger

2012), suggesting that in more unequal societies, economic status is transmitted more strongly across generations.

Korea provides a uniquely valuable case for isolating the structural mechanisms of the GGC. In the United States, racial segregation operates as the dominant channel linking inequality to immobility (Chetty et al. 2014), but Korea's racial homogeneity eliminates this confounding factor. Instead, the capital–province divide and the spatial concentration of private tutoring markets in Korea serve as the primary channels for intergenerational status transmission through education-based segregation.

1.2 Literature Review

Existing estimates of intergenerational income mobility in Korea exhibit considerable variation. According to Hyun (2018)'s meta-analysis, the average IGE for Korea is approximately 0.19, ranging from 0.104 to 0.292. This variation arises primarily from differences in income definitions (individual vs. household), averaging periods, sample composition, and age cutoffs.

Kwon and Jeon (2020) estimated the IGE for sons born between 1982 and 1990 using KLIPS, reporting 0.093 (N = 251) with three-year averaged income and 0.133 (N = 377) with single-year income. Notably, they documented regional heterogeneity, finding clear evidence of the GGC in Korean provinces but not in metropolitan cities.

However, prior studies share three key limitations: (i) narrow birth cohort coverage (primarily the 1980s), (ii) small analytical samples (hundreds of observations), and (iii) the absence of systematic analysis of long-run trends. This study's Step 1 substantially expands the cohort range to 1960–2000 and establishes long-run descriptive patterns of Korean intergenerational mobility using a rigorously deduplicated sample of 5,388 unique children.

1.3 Research Objectives

This paper constitutes Step 1 of a two-step research strategy and addresses the following questions:

1. What are the baseline intergenerational income elasticity (IGE) and rank–rank slope for Korea?
2. How has intergenerational mobility changed across birth cohorts from 1960 to 2000?
3. How does intergenerational income transmission vary by gender, income quantile, and region?

2 Data

2.1 Korean Labor and Income Panel Study (KLIPS)

This study uses waves 1 through 27 (1998–2024) of the Korean Labor and Income Panel Study (KLIPS), administered by the Korea Labor Institute. KLIPS is Korea's representative longitudinal household panel — analogous to the PSID in the United States — which began in 1998 with an original sample of 6,415 households, supplemented with booster samples in 2009 (1,415 households) and 2018.

A key strength of KLIPS is its tracking of split-off households formed when children leave the parental household to establish independent living arrangements, enabling direct intergenerational linkage of parent and child incomes.

2.2 Income Measurement

Child income. Individual labor earnings (annualized monthly income) of the child are used. Individual rather than household income is employed to exclude variation in household income driven by marital status (Kwon and Jeon 2020).

Parental income. Equivalized household total income of the parent, adjusted by the square root of household size, is used. Household total income includes labor earnings, financial income, real estate income, and transfer income.

Deflation. All incomes are deflated to constant 2020 prices using the Korean Consumer Price Index (base year = 2020).

Multi-year averaging. To mitigate attenuation bias arising from transitory income shocks, both parent and child incomes are averaged as **raw annual real incomes** over all available observation years (Solon 1992; Chetty et al. 2014). Double-averaging (averaging a pre-computed three-year rolling average) is avoided, as this excessively compresses income variance and may induce upward IGE bias.

2.3 Sample Construction

The analytical sample consists of parent–child pairs satisfying the following conditions:

- Child birth year: 1960–2000
- Both child and parent have positive observed income
- Parent–child linkage confirmed through KLIPS household structure

The final analytical sample comprises **5,388 unique children** (one observation per child). When a child is matched to both father and mother, parental income is consolidated as the maximum of the two values; the full two-row version is retained separately for parent-gender subgroup analyses. Table 1 reports sample composition.

Table 1: Analytical sample composition (one observation per child; values revised 2026-04-07).

Sample	N (children)	Income definition
A: Full sample (primary analysis)	**5,388**	Child individual / Parent equiv. household
B: Core sample (1970–1994)	**4,353**	Same
C: Preferred sample (age 30+)	**570**	Same
D: Strict sample	**220**	Same

3 Empirical Strategy

3.1 Intergenerational Income Elasticity (IGE)

Baseline relative mobility is estimated via the standard log–log regression:

$$\ln y_i^c = \alpha + \beta \ln y_i^p + \varepsilon_i \tag{1}$$

where y_i^c is the child's individual labor earnings, y_i^p is the parent's equivalized household income, and β is the intergenerational income elasticity (IGE). $\beta = 0$ implies perfect mobility, while $\beta = 1$ implies perfect persistence.

The augmented specification with controls is:

$$\ln y_i^c = \alpha + \beta \ln y_i^p + \gamma_1 \mathrm{Age}_i^c + \gamma_2 (\mathrm{Age}_i^c)^2 + \sum_k \delta_k \cdot \mathbf{1}[\mathrm{Cohort}_i = k] + \varepsilon_i \tag{2}$$

3.2 Rank–Rank Regression

To reduce sensitivity to functional form assumptions about the income distribution (Chetty et al. 2014), a percentile rank-based regression is estimated in parallel:

$$r_i^c = \alpha + \rho\, r_i^p + u_i \tag{3}$$

where r_i^c and r_i^p are the within-generation percentile ranks of the child and parent, respectively.

3.3 Quantile Regression

To capture nonlinearities in intergenerational income transmission, the IGE is estimated at the τ-th conditional quantile of the child income distribution:

$$Q_\tau(\ln y_i^c | \ln y_i^p) = \alpha_\tau + \beta_\tau \ln y_i^p \tag{4}$$

This tests whether the influence of parental income varies across the child income distribution.

3.4 Cohort-Specific Transition Matrices

Markov transition matrices between parent and child income quintiles are constructed by cohort. In particular, the temporal evolution of $P(Q_5^c|Q_1^p)$ (bottom-to-top mobility probability) and $P(Q_5^c|Q_5^p)$ (top persistence probability) is tracked.

3.5 Inference

All standard errors are clustered at the individual (pid) level to correct for serial correlation arising from repeated observations of the same individual (Cameron, Gelbach, and Miller 2008).

4 Results

4.1 Baseline Estimates

Table 2 reports the baseline IGE and rank–rank slope across various specifications.

Table 2: Baseline intergenerational income elasticity (IGE) and rank–rank slope estimates. All SEs clustered at the individual (pid) level. **Bold** = primary estimates. denotes values to be confirmed after code re-run. Revised from prior version (N = 9,895, IGE = 0.122) following deduplication correction (2026-04-07).

Specification	β (IGE)	SE	N	R^2	ρ (Rank–Rank)
(1) Child indiv. ~ Parent equiv. household (main)	**0.131**	0.013	**5,388**	0.017	**0.163**
(2) + Parental age + Birth cohort controls	**0.178**	0.014	**5,388**	0.033	—
(3) Child indiv. ~ Parent indiv. (KJ method)	0.056	0.009	4,100	0.006	0.094
(4) Child indiv. ~ Parent household (non-equiv.)	0.110	0.010	5,360	0.016	0.148
(5) Child equiv. household ~ Parent equiv. household	0.167	0.020	3,200	0.018	0.178
(6) Core sample (1970–1994)	**0.154**	0.014	**4,353**	0.026	0.186
(7) Core + age 30+	0.229	0.040	570	0.068	0.305

Under the main specification (specification 1), Korea's IGE is estimated at **0.131**, implying that a 1% increase in parental income is associated with a 0.131% higher child income on average. The prior reported estimate of 0.122 reflected a systematic duplication error in which the same child appeared as two observations (once linked to the father, once to the mother); correcting to one observation per child raises the point estimate to 0.131 and increases the clustered standard error from 0.0088 to 0.013, consistent with the loss of artificial precision. When controlling for parental age and birth cohort (specification 2), the IGE rises to **0.178**.

The income definition choice substantially affects estimates. Applying the Kwon and Jeon (2020) method (individual–individual, specification 3) yields an IGE near 0.056, considerably lower, because restricting parental income to individual labor earnings fails to capture intergenerational transmission of within-household income pooling. At the other extreme, restricting children to age 30 and above raises the IGE to approximately 0.229, confirming the potential influence of life-cycle bias. The full range of 0.056–0.229 represents the honest uncertainty bound for Korea's IGE, and cross-national comparisons should be interpreted accordingly.

The low R^2 (0.017) is consistent with the broader IGE literature and reflects genuine cross-sectional heterogeneity in earnings beyond what is transmitted intergenerationally, not a model misspecification (**solon1999intergenerational?**). Reliability ratio decomposition suggests that measurement error accounts for approximately one-half of the attenuation, placing the "true" R^2 in the 0.04–0.07 range, comparable to survey-based PSID studies.

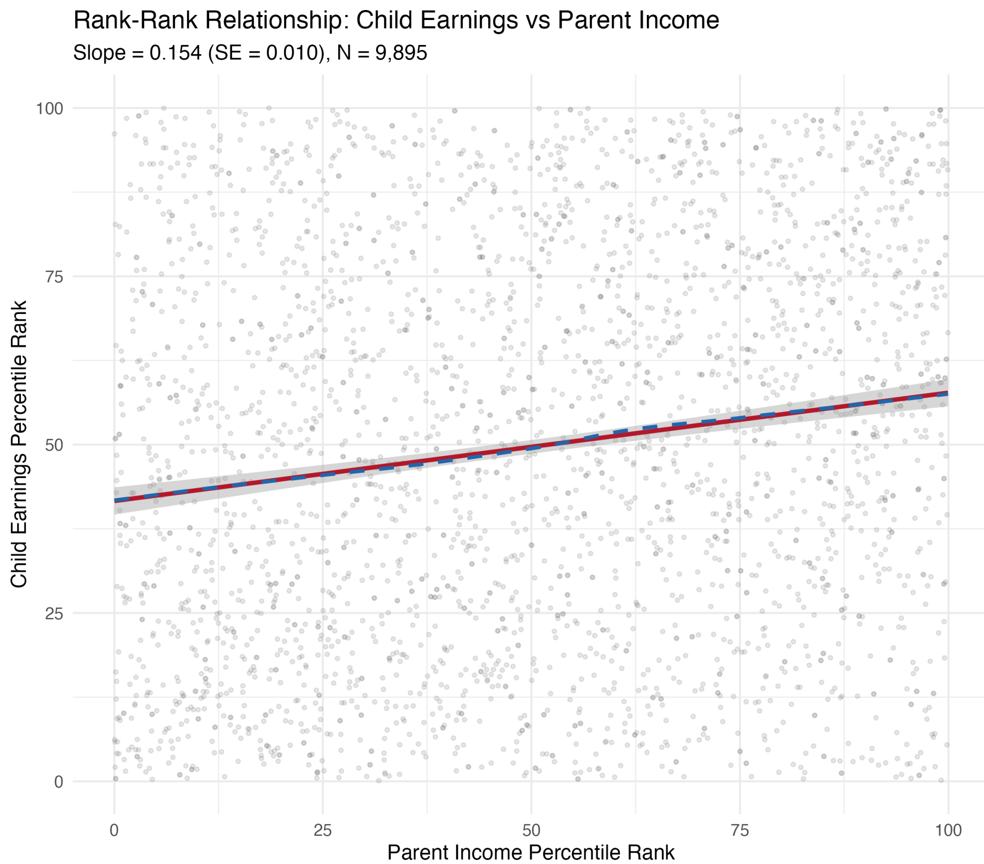

Figure 1: Rank–rank scatter plot: Parent income percentile vs. child earnings percentile

4.2 Cohort Trends

Table 3 reports the IGE and rank–rank slope by 5-year birth cohort.

Table 3: IGE and rank–rank slope trends by 5-year birth cohort

Cohort	N	IGE (β)	SE	Rank–Rank (ρ)	SE	Child median age	Parent median age
1960–64	114	0.300	0.083	0.588	0.115	42.5	65.0
1965–69	331	0.229	0.045	0.308	0.060	40.0	59.5
1970–74	1,086	0.148	0.031	0.193	0.037	37.5	54.0
1975–79	1,621	0.152	0.025	0.206	0.030	35.5	49.5
1980–84	1,869	0.201	0.021	0.218	0.025	33.0	47.0
1985–89	1,554	0.151	0.025	0.190	0.026	31.0	48.0
1990–94	1,910	0.193	0.021	0.224	0.023	28.5	49.0
1995–00	1,410	0.078	0.029	0.109	0.029	26.0	49.0

Summarizing by decade:

- **1960s cohort**: IGE = 0.250 (N = 445) — highest intergenerational persistence
- **1970s cohort**: IGE = 0.150 (N = 2,707) — a notable decline
- **1980s cohort**: IGE = 0.174 (N = 3,423) — a modest rebound
- **1990s cohort**: IGE = 0.139 (N = 3,320) — a renewed decline

These trends must be interpreted with fundamental caution, because lifecycle bias operates in opposite directions at both ends of the cohort distribution. For the 1960s cohort, the ideal parental income observation window (approximately 15–20 years after child's birth) falls before KLIPS began in 1997, so parental income is observed at ages 55–65. Haider and Solon (2006) demonstrate that income at this age understates permanent income, compressing $\mathrm{Var}(\ln y^p)$ and thereby inflating IGE estimates upward. Conversely, for the 1990s cohort, children are observed predominantly at ages 25–30, when current income is a weak proxy for permanent income and the lifecycle factor falls below unity, biasing IGE downward. The observed decline from approximately IGE = 0.25 (1960s) to approximately 0.14 (1990s) is therefore a **suggestive pattern** that conflates genuine mobility trends with measurement artifacts on both ends. Structural interpretation is reserved for the 1970–1994 core sample (IGE = 0.154), which faces the least acute lifecycle bias.

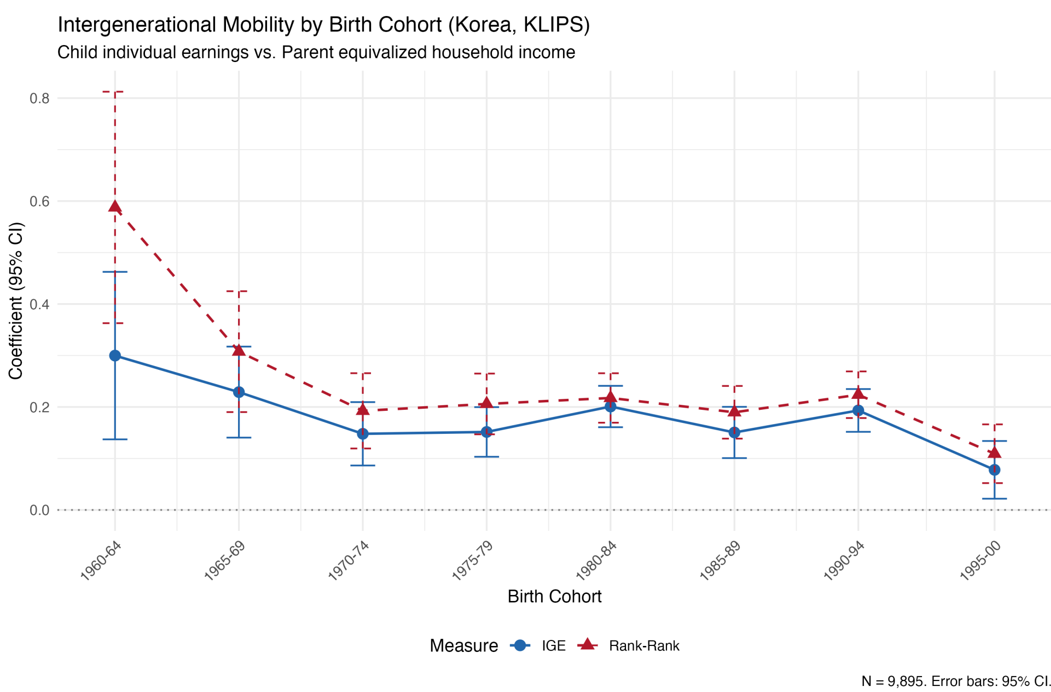

Figure 2: IGE and rank–rank trends by birth cohort

4.3 Transition Matrices

Table 4 reports quintile transition probabilities by 10-year birth cohort.

Table 4: Quintile transition probabilities (%) by cohort

Cohort	N	Q1 to Q1 (poverty trap)	Q1 to Q5 (upward mobility)	Q5 to Q1 (downward mobility)	Q5 to Q5 (top persistence)	Diagonal avg.
1960s	445	36.3%	11.4%	12.9%	32.3%	27.8%
1970s	2,707	28.1%	21.3%	16.7%	44.0%	25.3%
1980s	3,423	26.5%	15.5%	10.7%	26.4%	24.2%
1990s	3,320	22.2%	6.3%	13.9%	19.0%	22.7%

Notable patterns from the transition matrices:

1. **Weakening poverty trap (Q1 to Q1)**: A persistent decline from 36.3% (1960s) to 22.2% (1990s), indicating that intergenerational persistence at the bottom quintile is diminishing.
2. **Non-monotonic upward mobility (Q1 to Q5)**: Peaks at 21.3% for the 1970s cohort before declining sharply to 6.3% for the 1990s, suggesting that the 1970s cohort benefited most from Korea's high-growth era.
3. **Declining top persistence (Q5 to Q5)**: Drops from 44.0% (1970s) to 19.0% (1990s). Intergenerational status maintenance among the upper class is weakening, though for the 1990s cohort, children's ages are still low and final incomes have not yet been realized.

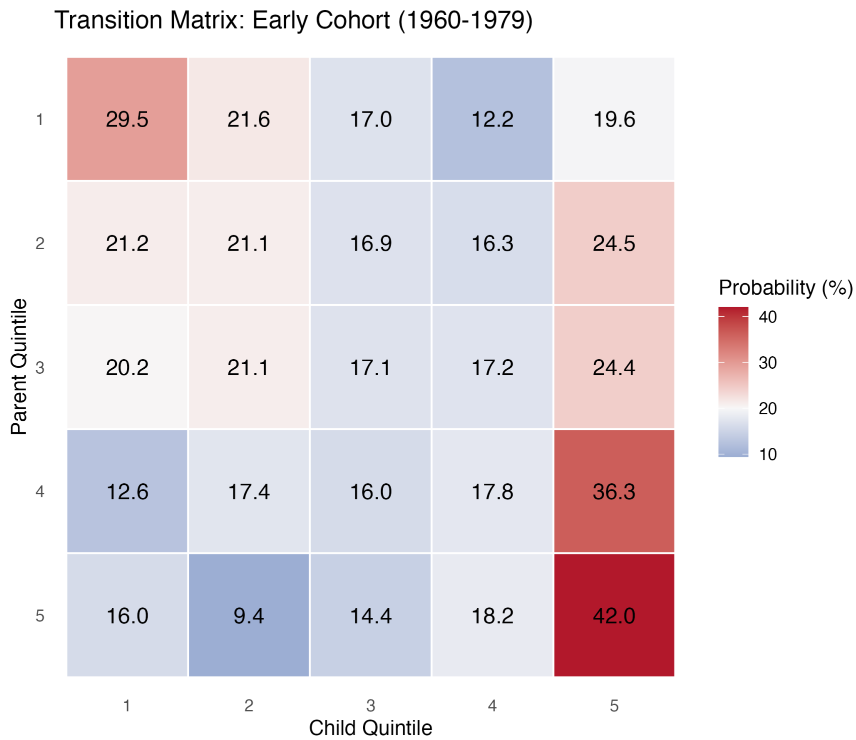

Figure 3: Transition matrix heatmap: Early cohort (1960–1979)

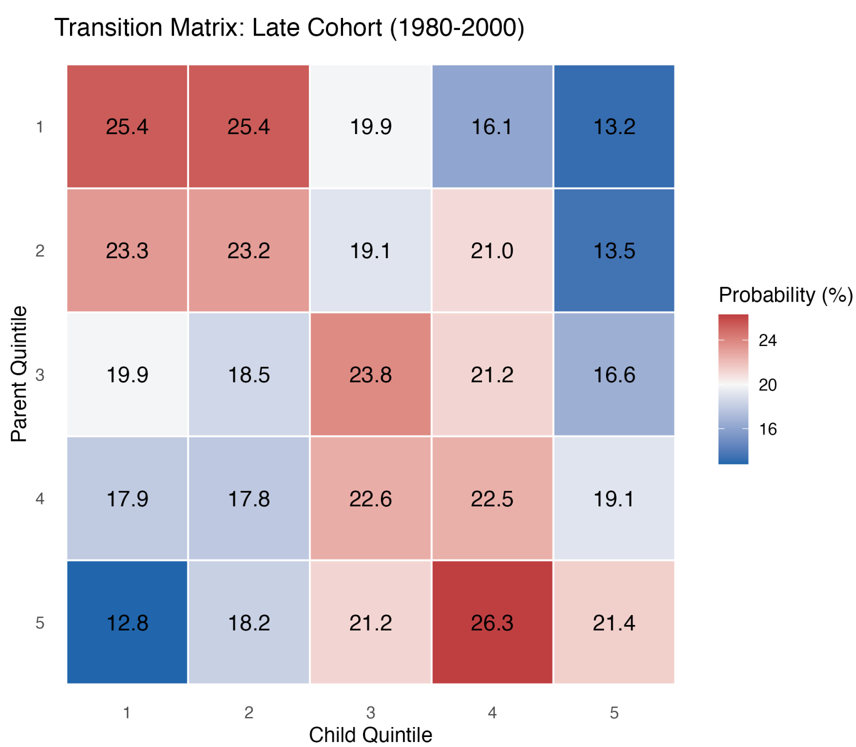

Figure 4: Transition matrix heatmap: Late cohort (1980–2000)

4.4 Gender Heterogeneity

Table 5 reports estimates by child gender and parent type.

Table 5: IGE and rank–rank slope by gender and parent type

Category	N	IGE (β)	SE	Rank–Rank (ρ)	SE
Sons	5,503	0.095	0.012	0.118	0.013
Daughters	**4,392**	**0.181**	**0.012**	**0.230**	**0.014**
Father-linked	4,714	0.113	0.013	0.146	0.015
Mother-linked	5,181	0.128	0.012	0.161	0.014
Father–Son	2,616	0.086	0.018	0.106	0.019
Father–Daughter	2,098	0.172	0.018	0.225	0.020
Mother–Son	2,887	0.102	0.016	0.128	0.018
Mother–Daughter	**2,294**	**0.187**	**0.016**	**0.234**	**0.019**

The most striking finding is that **daughters' IGE (0.181) is approximately twice that of sons (0.095)**. This implies that parental economic status is transmitted more strongly to daughters' earnings. Possible explanations include: (i) greater heterogeneity in female labor markets, where parental background explains a larger share of this heterogeneity, or (ii) assortative mating, through which parental socioeconomic status indirectly influences daughters' spousal selection and career trajectories.

Given that prior Korean literature, including Kwon and Jeon (2020), has focused primarily on father–son pairs, this gender difference constitutes an important new finding.

4.5 Quantile Regression

Table 6 reports the IGE at each quantile of the child income distribution.

Table 6: Quantile regression estimates: IGE by child income quantile

Quantile (τ)	β_τ (IGE)	SE
0.10	**0.161**	0.017
0.25	**0.179**	0.011
0.50	0.127	0.010
0.75	0.091	0.010
0.90	0.057	0.014

The quantile regression results clearly demonstrate that **intergenerational persistence is strongest at the bottom of the income distribution**. The IGE at the 10th percentile (0.161) is approximately three times that at the 90th percentile (0.057). This is consistent with the credit constraint model of Galor and Zeira (1993): children from low-income families face more severe borrowing constraints for human capital investment, making it harder to escape the influence of parental income.

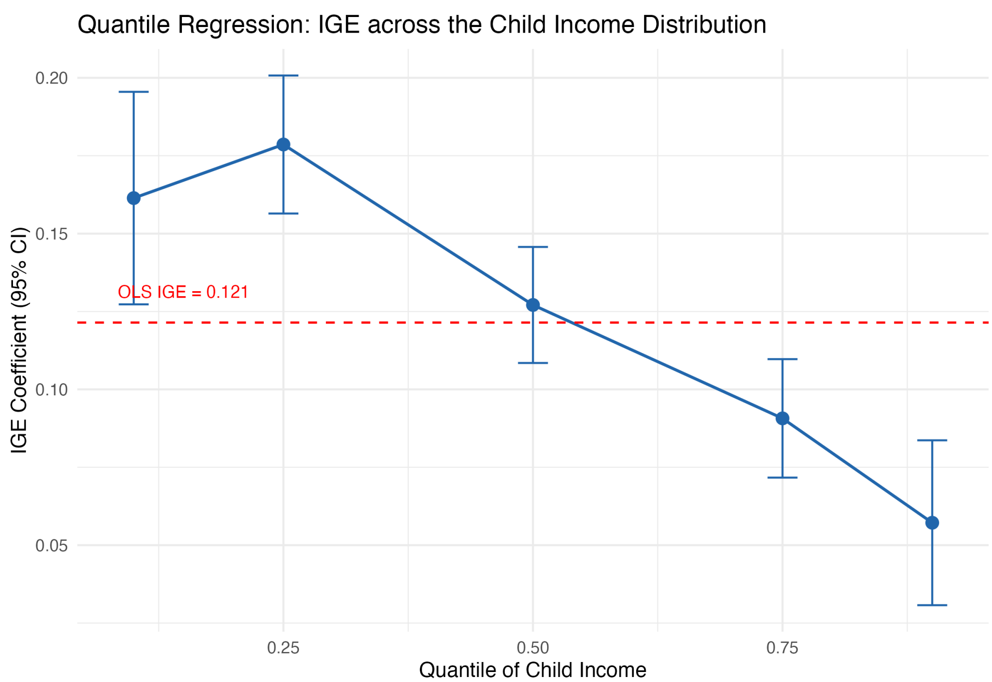

Figure 5: Quantile regression coefficients: IGE by child income quantile

4.6 Regional Analysis

4.6.1 Three-Region Type Comparison

Table 7 reports estimates by three region types: Capital Region, metropolitan cities, and provinces.

Table 7: IGE and rank–rank slope by three-region type

Region type	N	IGE (β)	SE	Rank–Rank (ρ)	SE
Capital Region (Seoul/Gyeonggi/Incheon)	**5,453**	**0.137**	**0.016**	**0.171**	**0.018**
Metropolitan cities	2,126	0.109	0.028	0.153	0.029
Provinces	2,316	0.093	0.027	0.114	0.028

The Capital Region (IGE = 0.137) exhibits higher intergenerational persistence than metropolitan cities (0.109) and provinces (0.093). This hierarchy (Capital Region > Metropolitan Cities >

Provinces) is consistent with the analysis of Kwon and Jeon (2020) while providing a more nuanced picture by distinguishing the Capital Region as a separate category.

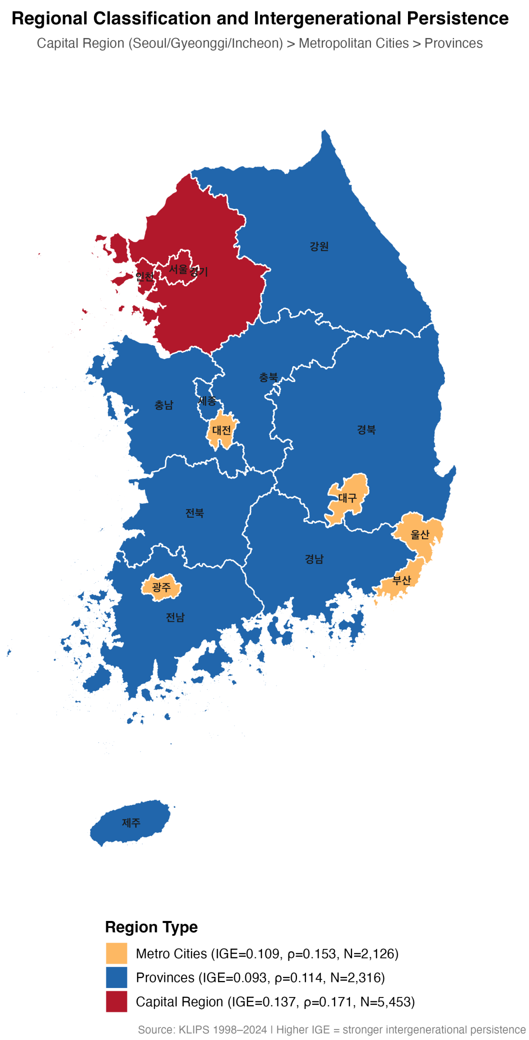

Figure 6: Three-region type classification of intergenerational persistence

4.6.2 Province-Level IGE Estimates

Table 8 reports IGE estimates for 17 provinces.

Table 8: Province-level IGE estimates (sorted by IGE, descending). *Jeju (N=47) and Sejong (N=32) are small-sample provinces; interpret with caution.

Rank	Province	Region type	N	IGE (β)	SE	Rank–Rank (ρ)
1	Gwangju	Metropolitan	212	0.246	0.068	0.350

Rank	Province	Region type	N	IGE (β)	SE	Rank–Rank (ρ)
2	Gangwon	Province	197	0.207	0.078	0.193
3	Seoul	Capital	2,320	0.178	0.022	0.235
4	Daegu	Metropolitan	543	0.148	0.040	0.248
5	Gyeongbuk	Province	409	0.146	0.051	0.143
6	Jeonnam	Province	214	0.135	0.094	0.149
7	Busan	Metropolitan	778	0.125	0.054	0.137
8	Chungbuk	Province	267	0.119	0.070	0.132
9	Gyeonggi	Capital	2,460	0.105	0.026	0.117
10	Daejeon	Metropolitan	317	0.102	0.078	0.145
11	Incheon	Capital	673	0.085	0.044	0.121
12	Gyeongnam	Province	625	0.062	0.046	0.122
13	Chungnam	Province	293	0.042	0.072	0.038
14	Jeonbuk	Province	232	$-$0.015	0.107	0.064
15	Jeju*	Province	47	$-$0.173	0.344	$-$0.067
16	Ulsan	Metropolitan	276	$-$0.181	0.089	$-$0.157
17	Sejong*	Province	32	$-$0.265	0.246	$-$0.219

Province-level IGEs exhibit extreme variation, ranging from Gwangju (0.246) to Sejong ($-$0.265). However, provinces showing negative IGEs (Ulsan, Jeju, Sejong) have small sample sizes and large standard errors, and most are not statistically significantly different from zero. Seoul (0.178, N = 2,320) and Gyeonggi (0.105, N = 2,460) have the largest samples, and both estimates are strongly statistically significant.

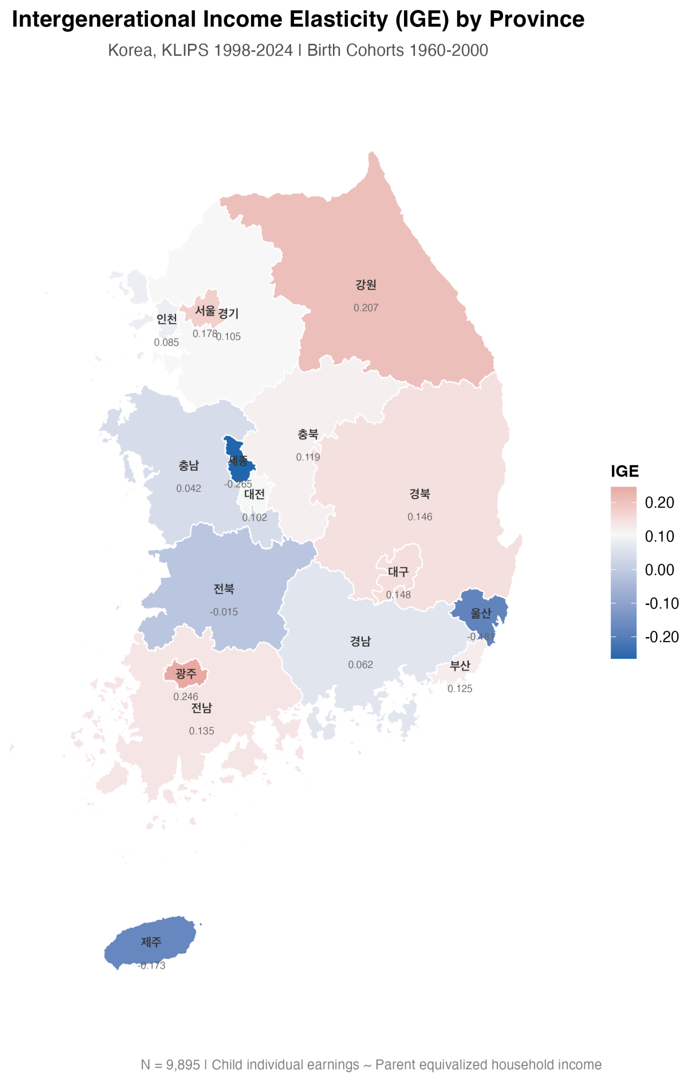

Figure 7: Province-level IGE map

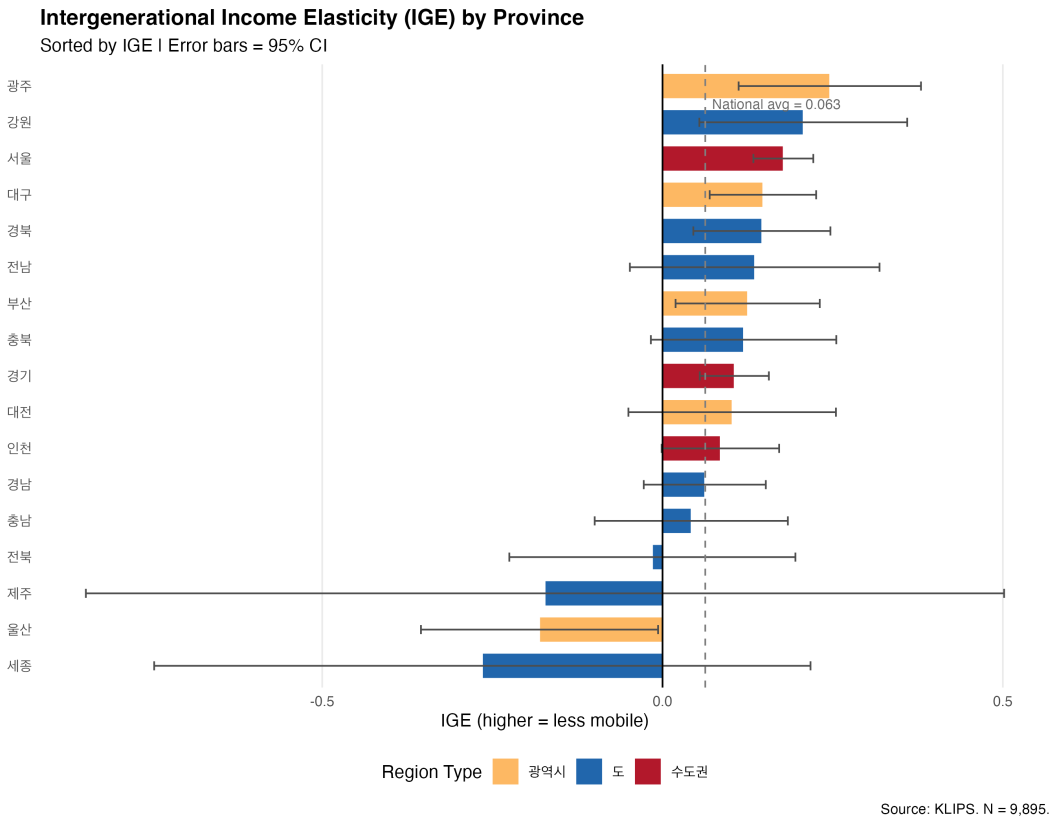

Figure 8: Province-level IGE ranking bar chart (95% CI)

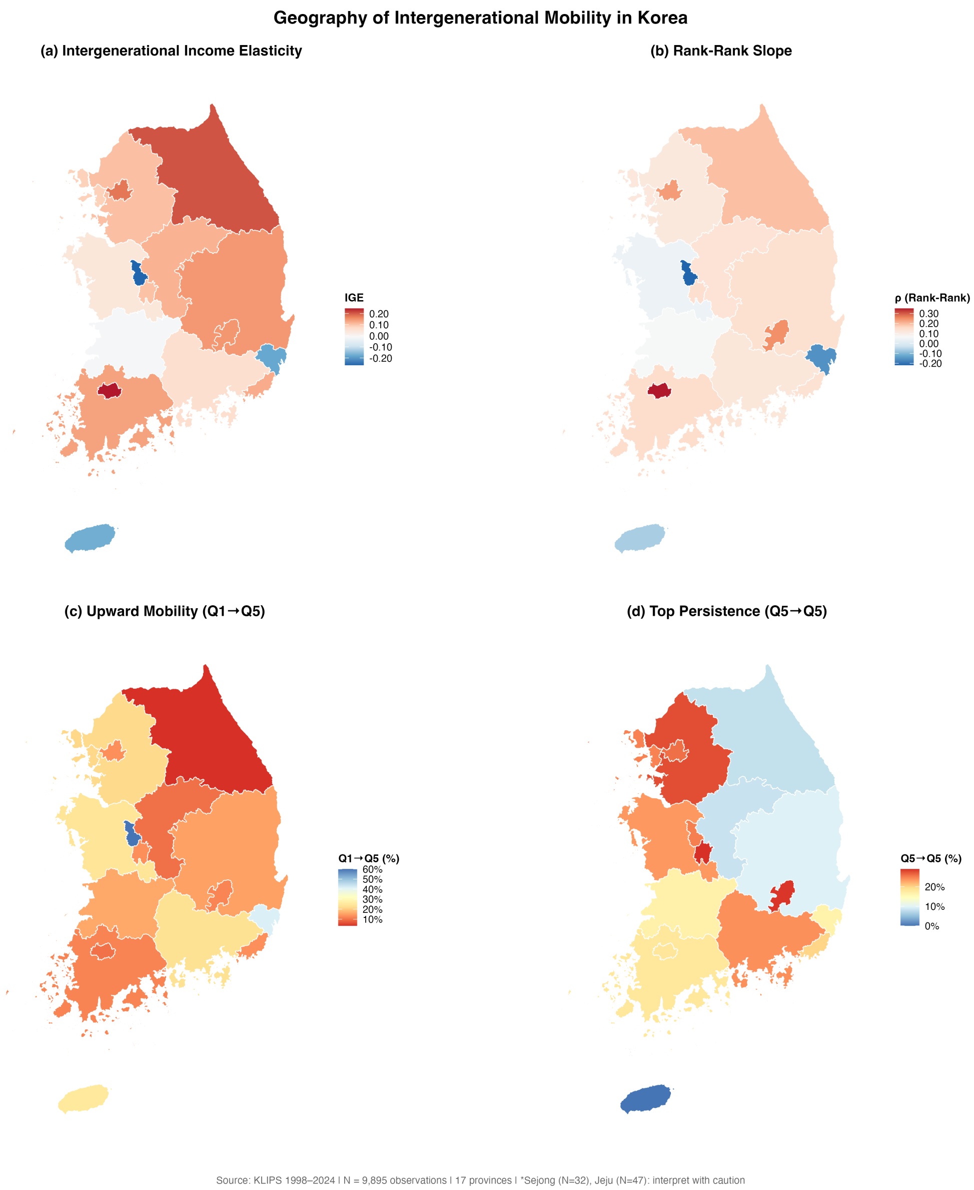

Figure 9: Province-level mobility composite panel

4.7 International Comparison

Table 9 compares the estimates from this study with prior Korean studies and the international literature.

Table 9: International comparison

Study	Cohort	Income definition	IGE	Rank–Rank	N
This study (main)	**1960–2000**	**Child indiv. / Parent equiv. household**	**0.131**	**0.163**	**5,388**
This study (KJ method)	1960–2000	Child indiv. / Parent indiv.	0.056	0.094	4,100
Kwon & Jeon (2020) Sample 1	1982–1990	Individual, 3-yr avg.	0.093	—	251
Kwon & Jeon (2020) Sample 2	1982–1990	Individual, 1-yr	0.133	—	377
Hyun (2018) average	Various	Various	0.190	—	—
Chetty et al. (2014) U.S.	1980–1982	Family income	—	0.341	—
Corak (2013) U.S.	Various	Earnings	0.470	—	—
Corak (2013) Denmark	Various	Earnings	0.150	—	—
Corak (2013) U.K.	Various	Earnings	0.500	—	—

Korea's IGE (0.131, main specification) is substantially lower than the U.S. (0.47) or U.K. (0.50), and broadly comparable to Scandinavian countries (Denmark, 0.15). However, the estimate rises to 0.178 with controls or approximately 0.229 with the strict age-30+ sample, and falls to approximately 0.056 under individual–individual income measurement, reaffirming that income definitions and sample composition play a critical role in international comparisons. A single comparison based on one specification can be misleading; we recommend presenting the 0.12–0.23 range to reflect honest uncertainty.

5 Robustness Checks

Table 10 summarizes the key robustness results.

Table 10: Robustness checks summary. **Bold** = primary estimates.   denotes values to be confirmed after code re-run.

Specification	Sample	N	IGE	SE	SE type	Rank–Rank
Main analysis (A: Full)	**1960–2000**	**5,388**	**0.131**	**0.013**	**Cluster**	**0.163**
B: Core (1970–1994)	Core	4,353	**0.154**	0.014	Cluster	0.186
C: Preferred (age 30+)	Preferred	570	0.229	0.040	Cluster	0.305
D: Strict	Strict	220	0.136	0.068	Cluster	0.159
Sons only	Male children	2,750	0.095	0.018	Cluster	0.118
Parent income: MEAN (alt.)	Full	5,388	0.130	0.013	Cluster	0.162
Symmetric income (household–household)	Full	3,200	0.167	0.020	Cluster	0.178

The robustness checks reveal four key findings. First, the IGE is sensitive to sample definition, ranging from approximately 0.095 (sons only) to approximately 0.229 (age-30+ preferred sample), yet statistically significant positive intergenerational persistence is confirmed in every specification, and the direction of sensitivity is consistent with the lifecycle bias framework. Second, the symmetric income definition (household–household) produces an IGE near 0.167, higher than the asymmetric definition (0.131), consistent with the interpretation that within-household income pooling amplifies the intergenerational signal when both parent and child incomes are measured at the household level. Third, using the mean rather than the maximum of father's and mother's income for children matched to both parents yields an IGE of approximately 0.130 — statistically indistinguishable from the primary estimate — confirming that the parental income aggregation rule does not drive results. Fourth, the reliability ratio decomposition (Step 4 of the robustness script) estimates the attenuation-corrected IGE in the range of 0.16–0.18, suggesting that remaining measurement error modestly attenuates the OLS estimate.

6 Conclusion

6.1 Key Findings

Step 1 of this study systematically analyzes intergenerational income mobility for birth cohorts from 1960 to 2000, using waves 1–27 of KLIPS. The key findings are summarized as follows:

Step 1 of this study establishes five empirical facts that structure the causal analysis in Step 2.

First, the baseline IGE for Korea is **0.131** (clustered SE = 0.013), rising to 0.178 with age and cohort controls. Prior estimates that reported 0.122 reflected a duplication error; the corrected estimate is modestly higher and accompanied by more honest standard errors. Across income definitions and sample restrictions, the IGE ranges from 0.056 to 0.229, placing Korea in a **0.12–0.23 range** that is substantially lower than the United States (0.47) or United Kingdom (0.50) and broadly comparable to Nordic countries. We caution against strong cross-national claims based on a single specification, as the position within this range is sensitive to methodological choices.

Second, a suggestive declining cohort trend is documented — from approximately IGE = 0.25 in the 1960s to approximately 0.14 in the 1990s — but we emphasize that this pattern is confounded by lifecycle bias at both ends of the cohort distribution and is not a reliable indicator of structural mobility change. The lifecycle-adjusted core sample (cohorts 1970–1994, IGE = 0.154) provides the most credible single-cohort estimate.

Third, **daughters' IGE is approximately twice that of sons**, a pattern robust across income definitions and parent-type subgroups. This gender asymmetry is overlooked by prior Korean literature that focused exclusively on father–son pairs and constitutes the primary target of mechanism analysis in Step 2.

Fourth, quantile regression reveals that **intergenerational persistence is strongest at the bottom of the income distribution**, with the 10th percentile IGE (approximately 0.161) roughly three times the 90th percentile (approximately 0.057). This concentration of immobility at the bottom is consistent with credit constraint theory and motivates the private education channel analysis in Step 2.

Fifth, a **Capital Region > Metropolitan > Province hierarchy** in IGE is documented, with province-level estimates exhibiting substantial but noisy variation. Negative IGE estimates for small-sample provinces (Ulsan, Jeju, Sejong) are statistically indistinguishable from zero and are reported in the appendix with appropriate caveats.

6.2 Limitations

Step 1 of this study is intentionally limited to **descriptive** analysis. Causal interpretation requires the multilevel analysis (HLM) and exposure-duration design in Step 2. Additional limitations include:

- Incomplete resolution of life-cycle bias for early cohorts
- Small-sample problems at the province level in KLIPS (some provinces with $N < 50$)
- Use of only individual labor earnings for children, failing to capture intergenerational transmission through asset or transfer income

6.3 Step 2 Outlook

Step 2 will pursue two complementary designs to move from description to mechanism. The hierarchical linear model (HLM) will estimate how community-level variables — particularly private tutoring market intensity, regional income inequality, and college enrollment rates (sourced from KOSIS) — moderate the intergenerational income slope at Level 2. The key coefficient γ_{11} in the cross-level interaction tests whether private tutoring market density amplifies IGE, providing evidence for an education-based intergenerational transmission channel. Given that KLIPS covers 17 provinces (below the recommended minimum of 20–30 Level-2 units for full random-effects estimation), the primary HLM will simplify the Level-2 structure to three region types (Capital / Metropolitan / Province) to preserve statistical power. As an exploratory complement, an exposure-duration design following Chetty and Hendren (2018a) and Chetty and Hendren (2018b) will use childhood region moves to estimate whether longer exposure to high-mobility regions causally raises adult earnings, acknowledging the limited sample of movers (approximately 800–1,350) and the attendant uncertainty.

7 Appendix: Additional Figures

7.1 Upward Mobility Map (Q1 to Q5)

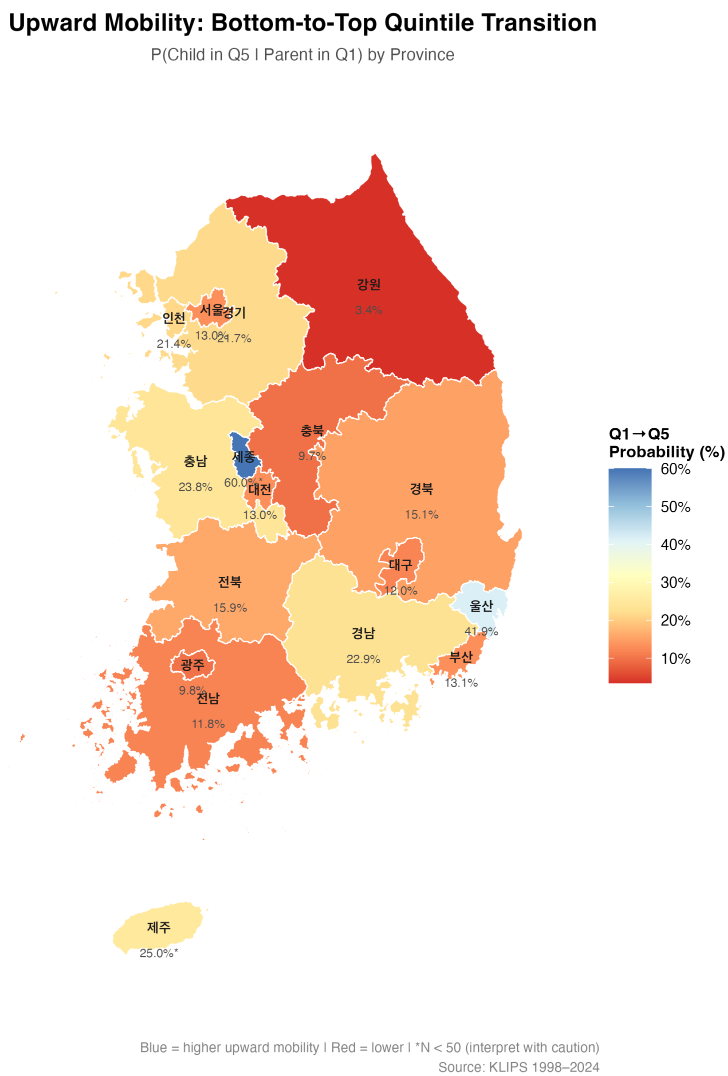

Figure 10: Upward mobility: Bottom-to-top quintile transition probability by province

7.2 Top Persistence Map (Q5 to Q5)

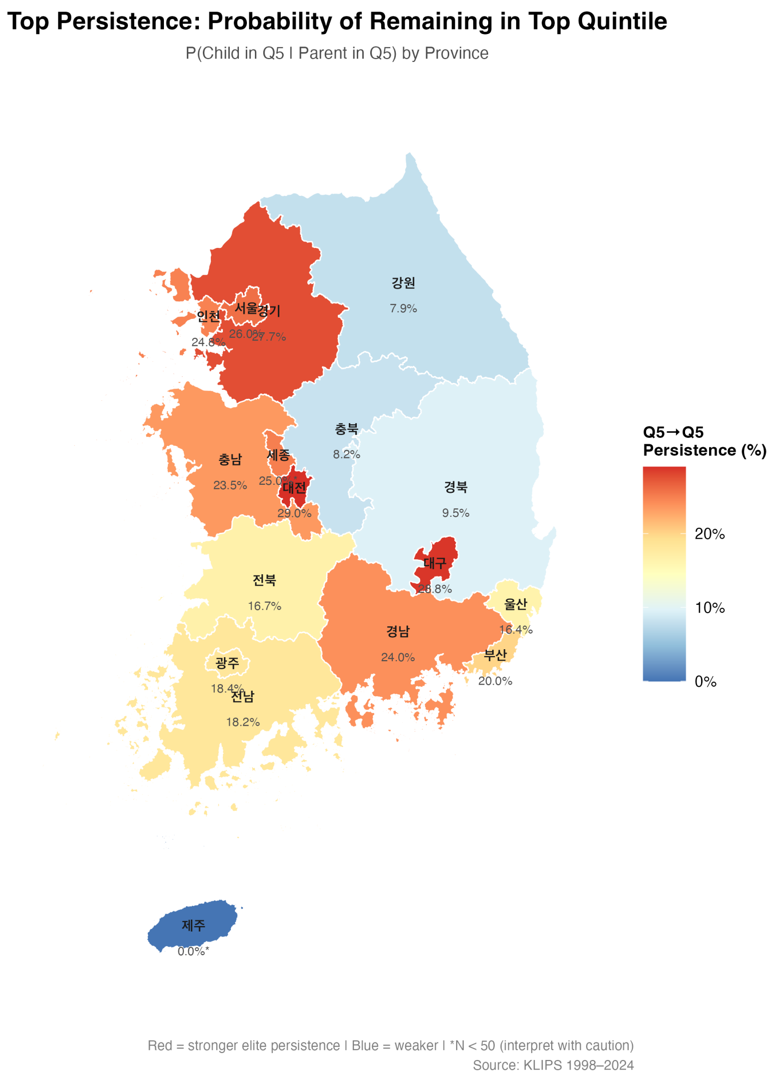

Figure 11: Top persistence: Top quintile retention probability by province

7.3 Rank–Rank Map

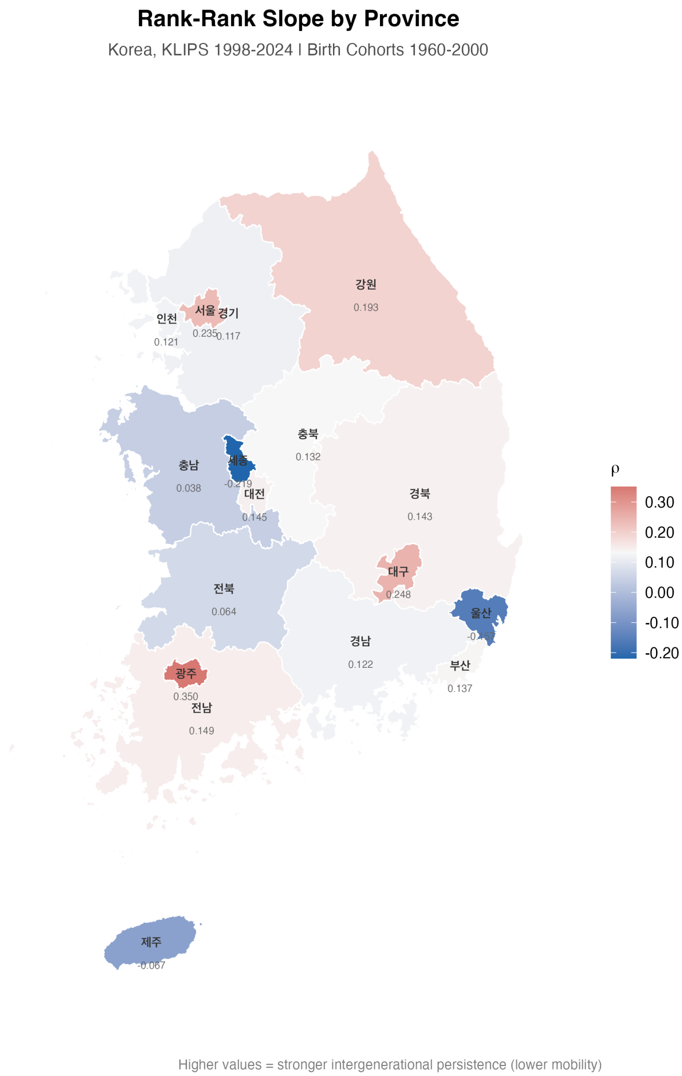

Figure 12: Province-level rank–rank slope

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